CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

BELO RANCHES

238 S BUHACH ROAD

MERCED CA 95342

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Belo Ranches

Physical address of dairy:

Number and Street	City	County	Zip Code
238 S Buhach RD	Merced	Merced	95342

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X215-X140-X005-XXXX

B. OPERATORS

Belo, Emanuel

Operator name: Belo, Emanuel

Telephone no.: Landline: ; Cellular: (209) 777-3145

Mailing Address Number and Street	City	State	Zip Code
511 Applegate RD	Atwater	CA	95301

This operator is responsible for paying permit fees.

C. OWNERS

Belo, Emanuel

Legal owner name: Belo, Emanuel

Telephone no.: Landline: ; Cellular: (209) 777-3145

Mailing Address Number and Street	City	State	Zip Code
511 Applegate RD	Atwater	CA	95301

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	490	1,097	561	11
Number under roof	0	0	0	0	0	0
Maximum number	0	0	490	1,097	561	11
Average number	0	0	490	1,097	561	11
Avg live weight (lbs)	0	0	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 17,923.35 *tons per reporting period*

Total nitrogen from manure: 179,835.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 125,884.85 *lbs per reporting period*

Total phosphorus from manure: 27,554.14 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 23,983,081 *gallons*

Total nitrogen generated: 50,561.05 *lbs*

Total phosphorus generated: 18,026.90 *lbs*

Total potassium generated: 231,165.97 *lbs*

Total salt generated: 990,440.77 *lbs*

	23,983,081	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	23,983,081	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	Separator solids *Imported SSB Manure*	4,914.00 *ton*	Dry-weight	70.9	14,900.00	7,100.00	7,100.00		0.00

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	42,613.23	20,305.63	20,305.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	42,613.23	20,305.63	20,305.63	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dickenson Ferry	38	38	2	manure	X065-X070-X053-XXXX X065-X070-X054-XXXX
Enos	20	20	2	both	X215-X140-X015-XXXX
F/D1/D2	50	50	2	both	X215-X140-X005-XXXX X215-X140-X016-XXXX
Gurr	88	88	2	manure	X215-X150-X006-XXXX
Middle	52	52	2	both	X215-X140-X006-XXXX X215-X140-X007-XXXX X215-X140-X016-XXXX
Totals for areas that were used for application	248	248	10		
Totals for areas that were not used for application					
Land application area totals	248	248	10		

B. CROPS AND HARVESTS

Dickenson Ferry

Field name: Dickenson Ferry

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	819.00 *ton*	Dry-weight		59.9	13,300.00	3,200.00	20,400.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	21.55	229.89	55.31	352.62	1,607.52

Dickenson Ferry

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	1,084.00 *ton*	Dry-weight		63.5	12,800.00	2,700.00	13,400.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.53	266.55	56.23	279.04	999.56

Enos

Field name: Enos

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	454.00 *ton*	Dry-weight		65.4	14,300.00	1,900.00	15,200.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	22.70	224.63	29.85	238.77	1,508.01

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	541.00 *ton*	Dry-weight		69.9	15,000.00	1,500.00	22,000.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.05	244.26	24.43	358.25	1,025.90

F/D1/D2

Field name: F/D1/D2

09/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 09/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	1,136.00 *ton*	Dry-weight		70.8	15,500.00	3,800.00	23,800.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	22.72	205.66	50.42	315.79	1,340.12

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,603.00 *ton*	Dry-weight		67.3	10,000.00	1,800.00	14,000.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.06	209.67	37.74	293.54	1,362.87

Gurr

Field name: Gurr

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 88 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	1,876.00 *ton*	Dry-weight		70.0	20,000.00	3,700.00	25,000.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	21.32	255.82	47.33	319.77	1,151.18

Gurr

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 88 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	2,543.00 *ton*	Dry-weight		65.1	13,000.00	2,900.00	12,400.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.90	262.22	58.49	250.12	1,028.70

Middle

Field name: Middle

09/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 09/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	1,123.00 *ton*	Dry-weight		67.3	13,400.00	3,000.00	25,100.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	21.60	189.26	42.37	354.51	1,384.14

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,480.00 *ton*	Dry-weight		69.6	15,500.00	1,700.00	22,800.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.46	268.22	29.42	394.55	1,176.71

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dickenson Ferry - 10/03/2022: Oats, silage-soft dough

Field name: Dickenson Ferry

Crop: Oats, silage-soft dough — Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	291.31	89.74	644.74	0.00	494.00 *ton*
Application event totals		291.31	89.74	644.74	0.00	

Dickenson Ferry - 05/26/2023: Corn, silage

Field name: Dickenson Ferry

Crop: Corn, silage — Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	338.20	161.16	161.16	0.00	1,482.00 *ton*
Application event totals		338.20	161.16	161.16	0.00	

06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.68	0.00	0.00	47.03	7,649,199.00 *gal*
Application event totals		1.68	0.00	0.00	47.03	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.28	0.00	0.00	35.74	5,812,427.00 *gal*
Application event totals		1.28	0.00	0.00	35.74	

Dickenson Ferry - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.82	0.00	0.00	22.85	3,716,890.00 *gal*
Application event totals		0.82	0.00	0.00	22.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.88	0.00	0.00	24.67	4,011,503.00 *gal*
Application event totals		0.88	0.00	0.00	24.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.56	0.00	0.00	43.59	7,088,977.00 *gal*
Application event totals		1.56	0.00	0.00	43.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.35	0.00	0.00	37.76	6,140,608.00 *gal*
Application event totals		1.35	0.00	0.00	37.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	36.34	5,909,219.00 *gal*
Application event totals		1.30	0.00	0.00	36.34	

Enos - 10/03/2022: Oats, silage-soft dough

Field name: Enos

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Enos - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.51	22.87	292.79	1,302.02	792,000.00 *gal*
Application event totals		98.51	22.87	292.79	1,302.02	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.31	18.61	321.87	1,685.36	792,000.00 *gal*
Irrigation Well	Ground water	9.59	0.00	0.00	0.00	6,048,000.00 *gal*
Application event totals		107.90	18.61	321.87	1,685.36	

04/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.63	29.47	400.16	1,622.27	648,000.00 *gal*
Application event totals		94.63	29.47	400.16	1,622.27	

Enos - 05/18/2023: Corn, silage

Field name: Enos

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	138.13	47.87	27.35	11,159.62	260.00 *ton*
Application event totals		138.13	47.87	27.35	11,159.62	

Enos - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.07	0.00	0.00	0.00	1,935,360.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.28	1,050,894.00 *gal*
Application event totals		3.51	0.00	0.00	12.28	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.07	0.00	0.00	0.00	1,935,360.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.23	1,046,406.00 *gal*
Application event totals		3.51	0.00	0.00	12.23	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.33	29.71	348.49	1,413.94	684,590.00 *gal*
Irrigation Well	Ground water	3.20	0.00	0.00	0.00	2,016,000.00 *gal*
Canal	Surface water	0.40	0.00	0.00	11.30	966,908.00 *gal*
Application event totals		57.93	29.71	348.49	1,425.24	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.71	0.00	0.00	0.00	2,338,560.00 *gal*
Canal	Surface water	0.33	0.00	0.00	9.14	782,000.00 *gal*
Application event totals		4.03	0.00	0.00	9.14	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.33	29.71	348.49	1,413.94	684,590.00 *gal*
Irrigation Well	Ground water	3.58	0.00	0.00	0.00	2,257,920.00 *gal*
Canal	Surface water	0.24	0.00	0.00	6.65	569,524.00 *gal*
Application event totals		58.15	29.71	348.49	1,420.60	

Enos - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.20	0.00	0.00	0.00	2,016,000.00 *gal*
Canal	Surface water	0.32	0.00	0.00	9.05	774,894.00 *gal*
Application event totals		3.52	0.00	0.00	9.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.33	29.71	348.49	1,413.94	684,590.00 *gal*
Irrigation Well	Ground water	3.20	0.00	0.00	0.00	2,016,000.00 *gal*
Canal	Surface water	0.33	0.00	0.00	9.21	788,144.00 *gal*
Application event totals		57.86	29.71	348.49	1,423.15	

F/D1/D2 - 09/30/2022: Oats, silage-soft dough

Field name: F/D1/D2

Crop: Oats, silage-soft dough

Plant date: 09/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.56	20.79	266.17	1,183.65	1,800,000.00 *gal*
Application event totals		89.56	20.79	266.17	1,183.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.37	16.91	292.61	1,532.14	1,800,000.00 *gal*
Irrigation Well	Ground water	9.61	0.00	0.00	0.00	12,257,280.00 *gal*
Application event totals		98.99	16.91	292.61	1,532.14	

F/D1/D2 - 09/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.12	26.20	355.70	1,442.02	1,440,000.00 *gal*
Application event totals		84.12	26.20	355.70	1,442.02	

F/D1/D2 - 05/18/2023: Corn, silage

Field name: F/D1/D2

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	138.13	47.87	27.35	11,159.62	650.00 *ton*
Application event totals		138.13	47.87	27.35	11,159.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.97	0.00	0.00	0.00	3,790,080.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.28	2,627,235.00 *gal*
Application event totals		3.41	0.00	0.00	12.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.04	0.00	0.00	0.00	3,870,720.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.23	2,616,015.00 *gal*
Application event totals		3.47	0.00	0.00	12.23	

F/D1/D2 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.15	23.05	270.38	1,097.03	1,327,869.00 *gal*
Irrigation Well	Ground water	3.23	0.00	0.00	0.00	4,112,640.00 *gal*
Canal	Surface water	0.40	0.00	0.00	11.30	2,417,270.00 *gal*
Application event totals		45.78	23.05	270.38	1,108.32	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.11	0.00	0.00	0.00	5,241,600.00 *gal*
Canal	Surface water	0.33	0.00	0.00	9.14	1,954,999.00 *gal*
Application event totals		4.44	0.00	0.00	9.14	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.28	22.02	258.36	1,048.27	1,268,852.00 *gal*
Irrigation Well	Ground water	4.24	0.00	0.00	0.00	5,402,880.00 *gal*
Canal	Surface water	0.24	0.00	0.00	6.65	1,423,809.00 *gal*
Application event totals		44.75	22.02	258.36	1,054.92	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.05	0.00	0.00	0.00	5,160,960.00 *gal*
Canal	Surface water	0.32	0.00	0.00	9.05	1,937,235.00 *gal*
Application event totals		4.37	0.00	0.00	9.05	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.47	20.49	240.34	975.13	1,180,328.00 *gal*
Irrigation Well	Ground water	4.05	0.00	0.00	0.00	5,160,960.00 *gal*
Canal	Surface water	0.33	0.00	0.00	9.21	1,970,359.00 *gal*
Application event totals		41.85	20.49	240.34	984.34	

Gurr - 10/05/2022: Oats, silage-soft dough

Field name: Gurr

Crop: Oats, silage-soft dough — Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	291.31	89.74	644.74	0.00	1,144.00 *ton*
Application event totals		291.31	89.74	644.74	0.00	

Gurr - 05/25/2023: Corn, silage

Field name: Gurr

Crop: Corn, silage — Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	338.20	161.16	161.16	0.00	3,432.00 *ton*
Application event totals		338.20	161.16	161.16	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.44	0.00	0.00	68.37	25,747,730.00 *gal*
Application event totals		2.44	0.00	0.00	68.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.58	0.00	0.00	44.30	16,685,754.00 *gal*
Application event totals		1.58	0.00	0.00	44.30	

Gurr - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.93	0.00	0.00	26.14	9,844,461.00 *gal*
Application event totals		0.93	0.00	0.00	26.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	38.54	14,513,305.00 *gal*
Application event totals		1.38	0.00	0.00	38.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	35.08	13,211,660.00 *gal*
Application event totals		1.25	0.00	0.00	35.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	33.70	12,690,220.00 *gal*
Application event totals		1.20	0.00	0.00	33.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	22.73	8,559,197.00 *gal*
Application event totals		0.81	0.00	0.00	22.73	

Middle - 09/30/2022: Oats, silage-soft dough

Field name: Middle

Crop: Oats, silage-soft dough

Plant date: 09/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Middle - 09/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.11	19.99	255.93	1,138.13	1,800,000.00 *gal*
Application event totals		86.11	19.99	255.93	1,138.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.62	14.31	247.59	1,296.43	1,584,000.00 *gal*
Irrigation Well	Ground water	9.55	0.00	0.00	0.00	12,660,480.00 *gal*
Application event totals		85.17	14.31	247.59	1,296.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.93	26.45	359.12	1,455.88	1,512,000.00 *gal*
Application event totals		84.93	26.45	359.12	1,455.88	

Middle - 05/18/2023: Corn, silage

Field name: Middle

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	138.13	47.87	27.35	11,159.62	676.00 *ton*
Application event totals		138.13	47.87	27.35	11,159.62	

Middle - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.68	0.00	0.00	0.00	3,548,160.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.28	2,732,324.00 *gal*
Application event totals		3.11	0.00	0.00	12.28	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	2.80	0.00	0.00	0.00	3,709,440.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.23	2,720,656.00 *gal*
Application event totals		3.23	0.00	0.00	12.23	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.89	33.29	390.55	1,584.59	1,994,754.00 *gal*
Irrigation Well	Ground water	2.98	0.00	0.00	0.00	3,951,360.00 *gal*
Canal	Surface water	0.40	0.00	0.00	11.30	2,513,961.00 *gal*
Application event totals		64.27	33.29	390.55	1,595.89	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.83	0.00	0.00	0.00	5,080,320.00 *gal*
Canal	Surface water	0.33	0.00	0.00	9.14	2,033,199.00 *gal*
Application event totals		4.16	0.00	0.00	9.14	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.89	33.29	390.55	1,584.59	1,994,754.00 *gal*
Irrigation Well	Ground water	4.09	0.00	0.00	0.00	5,421,600.00 *gal*
Canal	Surface water	0.24	0.00	0.00	6.65	1,480,761.00 *gal*
Application event totals		65.21	33.29	390.55	1,591.25	

Middle - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.77	0.00	0.00	0.00	4,999,680.00 *gal*
Canal	Surface water	0.32	0.00	0.00	9.05	2,014,724.00 *gal*
Application event totals		4.09	0.00	0.00	9.05	

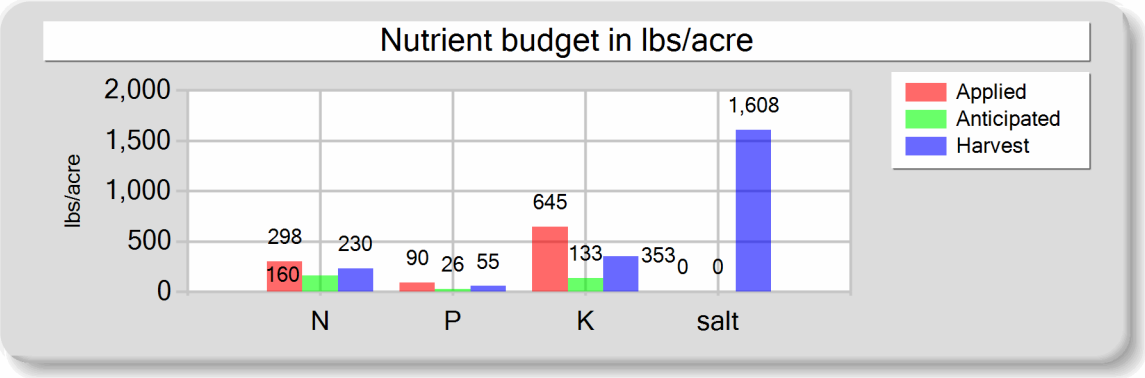
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.89	33.29	390.55	1,584.59	1,994,754.00 *gal*
Irrigation Well	Ground water	3.77	0.00	0.00	0.00	4,999,680.00 *gal*
Canal	Surface water	0.33	0.00	0.00	9.21	2,049,174.00 *gal*
Application event totals		64.99	33.29	390.55	1,593.80	

B. NUTRIENT BUDGET

Dickenson Ferry - 10/03/2022: Oats, silage-soft dough

Field name: Dickenson Ferry Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	291.31	89.74	644.74	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.31	89.74	644.74	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	229.89	55.31	352.62	1,607.52
Nutrient balance	68.41	34.43	292.12	-1,607.52
Applied to removed ratio	1.30	1.62	1.83	0.00

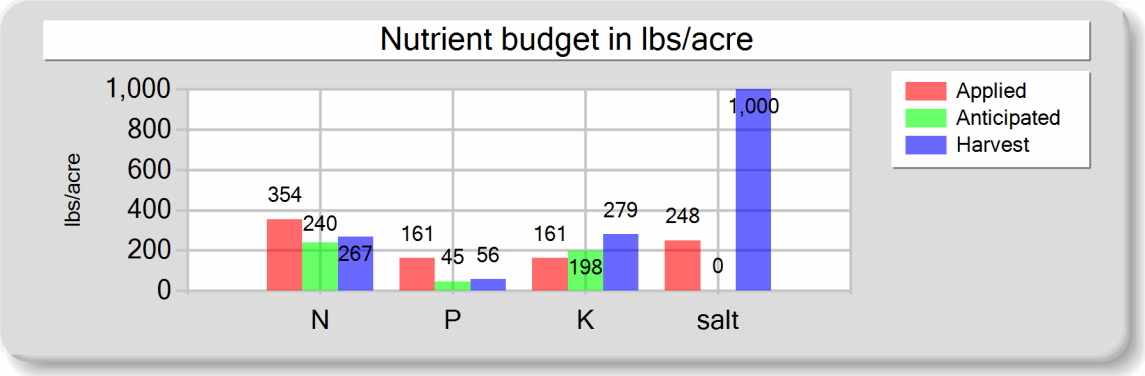
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Dickenson Ferry - 05/26/2023: Corn, silage

Field name: Dickenson Ferry Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	338.20	161.16	161.16	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.86	0.00	0.00	247.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	354.06	161.16	161.16	247.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	266.55	56.23	279.04	999.56
Nutrient balance	87.51	104.93	-117.89	-751.58
Applied to removed ratio	1.33	2.87	0.58	0.25

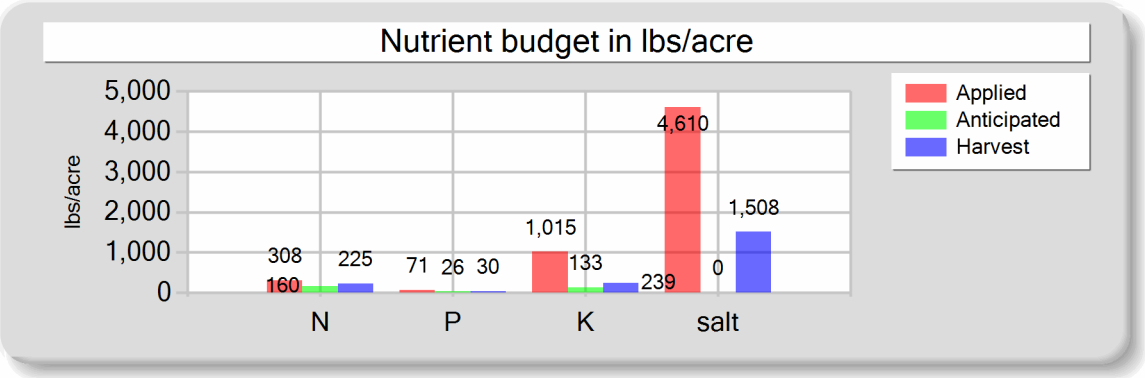
Fresh water applied	
40,328,823.00	*gallons*
1,485.17	*acre-inches*
39.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Enos - 10/03/2022: Oats, silage-soft dough

Field name: Enos | Crop: Oats, silage-soft dough | Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	291.46	70.94	1,014.82	4,609.64
Fresh water	9.59	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.04	70.94	1,014.82	4,609.64
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	224.63	29.85	238.77	1,508.01
Nutrient balance	83.41	41.10	776.05	3,101.64
Applied to removed ratio	1.37	2.38	4.25	3.06

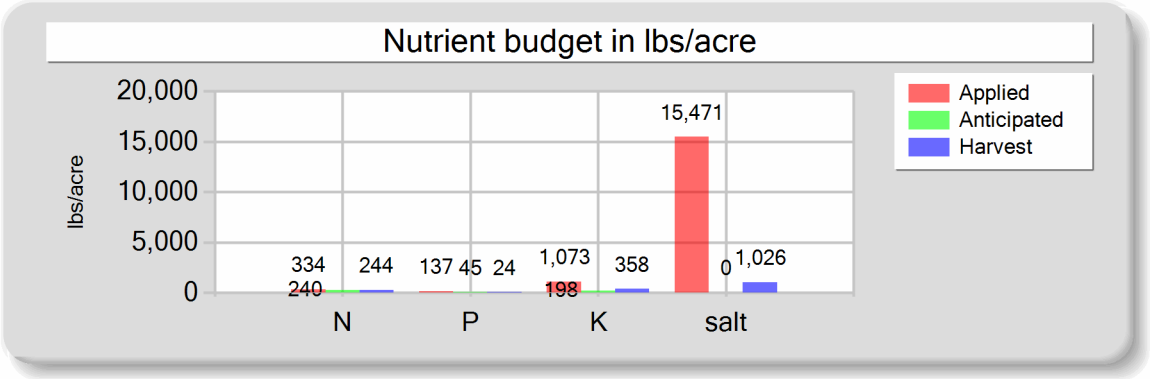
Fresh water applied	
6,048,000.00	*gallons*
222.73	*acre-inches*
11.14	*inches/acre*

Process wastewater applied	
2,232,000.00	*gallons*
82.20	*acre-inches*
4.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

Enos - 05/18/2023: Corn, silage

Field name: Enos | Crop: Corn, silage | Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	138.13	47.87	27.35	11,159.62
Process wastewater	162.99	89.12	1,045.46	4,241.83
Fresh water	25.51	0.00	0.00	69.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	333.63	136.99	1,072.81	15,471.30
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	244.26	24.43	358.25	1,025.90
Nutrient balance	89.36	112.56	714.56	14,445.40
Applied to removed ratio	1.37	5.61	2.99	15.08

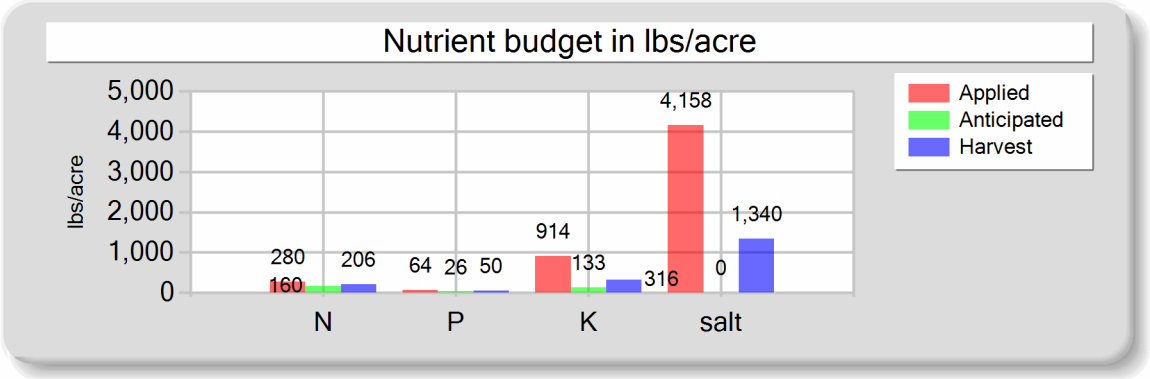
Fresh water applied	
20,493,970.00	*gallons*
754.72	*acre-inches*
37.74	*inches/acre*

Process wastewater applied	
2,053,770.00	*gallons*
75.63	*acre-inches*
3.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

F/D1/D2 - 09/30/2022: Oats, silage-soft dough

Field name: F/D1/D2 Crop: Oats, silage-soft dough Plant date: 09/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.05	63.90	914.48	4,157.81
Fresh water	9.61	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.66	63.90	914.48	4,157.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	205.66	50.42	315.79	1,340.12
Nutrient balance	74.00	13.48	598.69	2,817.70
Applied to removed ratio	1.36	1.27	2.90	3.10

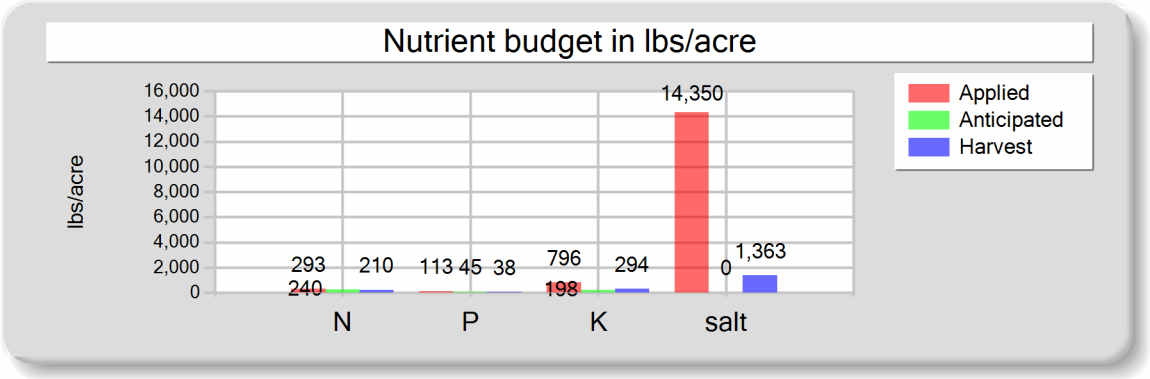
Fresh water applied	
12,257,280.00	*gallons*
451.39	*acre-inches*
9.03	*inches/acre*

Process wastewater applied	
5,040,000.00	*gallons*
185.61	*acre-inches*
3.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

F/D1/D2 - 05/18/2023: Corn, silage

Field name: F/D1/D2 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	138.13	47.87	27.35	11,159.62
Process wastewater	119.90	65.56	769.08	3,120.43
Fresh water	28.18	0.00	0.00	69.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.20	113.43	796.43	14,349.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.67	37.74	293.54	1,362.87
Nutrient balance	83.53	75.69	502.89	12,987.02
Applied to removed ratio	1.40	3.01	2.71	10.53

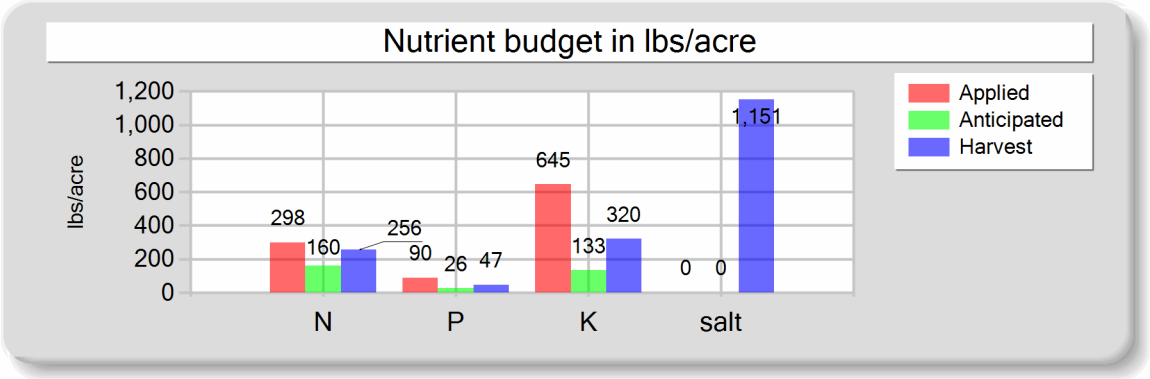
Fresh water applied	
47,686,762.00	*gallons*
1,756.14	*acre-inches*
35.12	*inches/acre*

Process wastewater applied	
3,777,049.00	*gallons*
139.10	*acre-inches*
2.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

Gurr - 10/05/2022: Oats, silage-soft dough

Field name: Gurr | Crop: Oats, silage-soft dough | Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	291.31	89.74	644.74	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.31	89.74	644.74	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	255.82	47.33	319.77	1,151.18
Nutrient balance	42.49	42.41	324.97	-1,151.18
Applied to removed ratio	1.17	1.90	2.02	0.00

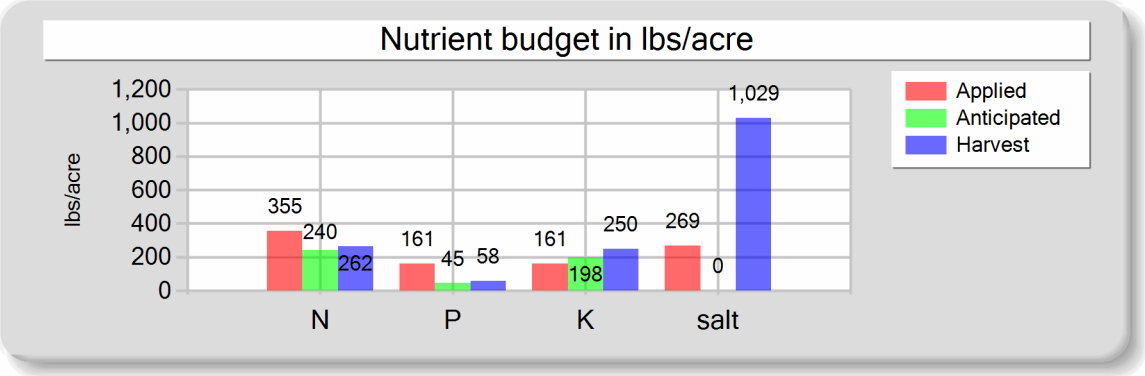
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Gurr - 05/25/2023: Corn, silage

Field name: Gurr | Crop: Corn, silage | Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	338.20	161.16	161.16	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.60	0.00	0.00	268.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	354.80	161.16	161.16	268.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	262.22	58.49	250.12	1,028.70
Nutrient balance	92.58	102.66	-88.96	-759.85
Applied to removed ratio	1.35	2.76	0.64	0.26

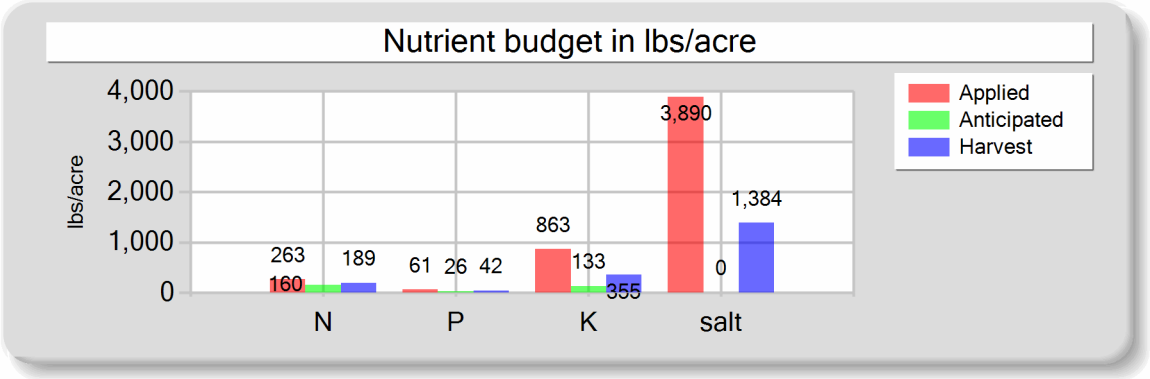
Fresh water applied	
101,252,327.00	*gallons*
3,728.78	*acre-inches*
42.37	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Middle - 09/30/2022: Oats, silage-soft dough

Field name: Middle | Crop: Oats, silage-soft dough | Plant date: 09/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.66	60.75	862.64	3,890.44
Fresh water	9.55	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.21	60.75	862.64	3,890.44
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	189.26	42.37	354.51	1,384.14
Nutrient balance	73.95	18.38	508.13	2,506.30
Applied to removed ratio	1.39	1.43	2.43	2.81

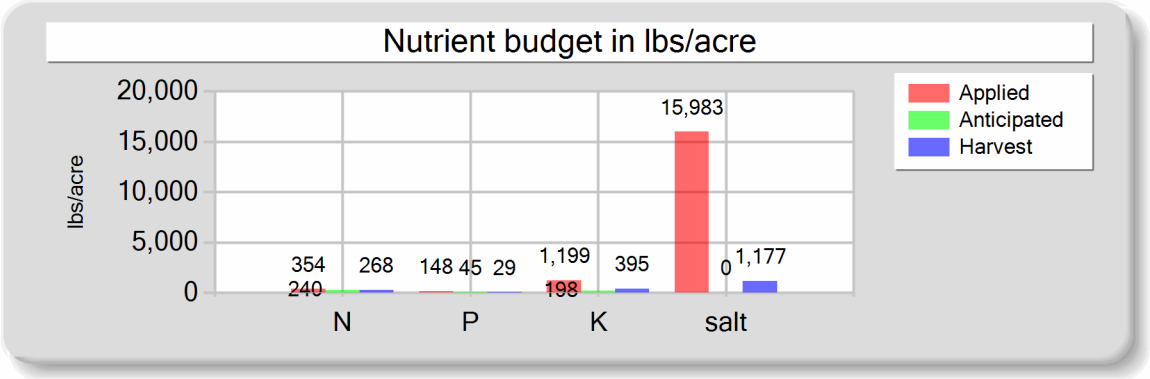
Fresh water applied	
12,660,480.00	*gallons*
466.24	*acre-inches*
8.97	*inches/acre*

Process wastewater applied	
4,896,000.00	*gallons*
180.30	*acre-inches*
3.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Middle - 05/18/2023: Corn, silage

Field name: Middle | Crop: Corn, silage | Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	138.13	47.87	27.35	11,159.62
Process wastewater	182.66	99.88	1,171.64	4,753.78
Fresh water	26.41	0.00	0.00	69.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	354.20	147.74	1,198.99	15,983.24
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	268.22	29.42	394.55	1,176.71
Nutrient balance	85.98	118.33	804.44	14,806.53
Applied to removed ratio	1.32	5.02	3.04	13.58

Fresh water applied	
47,255,039.00	*gallons*
1,740.24	*acre-inches*
33.47	*inches/acre*

Process wastewater applied	
5,984,262.00	*gallons*
220.38	*acre-inches*
4.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall Corral Manure

Sample and source description: Fall Corral Manure

Sample date: 10/29/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	6,500.00	46,700.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	3,500.00	2,000.00	9,200.00	5,500.00	2,000.00	1,500.00	3,000.00		81.60
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

Imported SSB Manure

Sample and source description: Imported SSB Manure

Sample date: 12/12/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	7,100.00	7,100.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022

Sample and source description: Lagoon Q4 2022

Sample date: 12/02/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	297.00	170.00	0.00	1.10	69.20	886.00								7,340.00	3,940
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	296.00	176.00	0.00	1.50	56.30	974.00								7,700.00	5,100
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	349.00	157.00	0.00	1.00	109.00	1,480.00								9,920.00	6,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	189.00	69.10	0.00	1.20	104.00	1,220.00								8,070.00	4,950
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	136.00	83.20	0.00	0.60	59.90	494.00								2,480.00	4,140
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

MID Canal

Sample description: MID Canal

Sample date: 10/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00								32.20	28
DL	0.01	0.01	0.01								10.00	1

Domestic Well

Domestic Well Milk Barn

Sample description: Domestic Well Milk Barn

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.20	0.00	10.20								1,100.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

Irrigation Well Dairy

Sample description: Irrigation Well Dairy

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.70	0.00	4.70								542.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well Enos

Sample description: Irrigation Well Enos

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.80	0.00	3.80								448.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well Middle

Sample description: Irrigation Well Middle

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.70	0.00	4.70								534.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Dickenson Ferry - 10/03/2022: Oats, silage-soft dough

Dickenson Ferry - 10/03/2022: Oats, silage-soft dough

Dickenson Ferry Oat Silage

Sample and source description: Dickenson Ferry Oat Silage

Sample date: 05/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	3,200.00	20,400.00		9.30
DL	0.01	0.01	0.01		1.00

Dickenson Ferry - 05/26/2023: Corn, silage

Dickenson Ferry Corn Silage

Sample and source description: Dickenson Ferry Corn Silage

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	2,700.00	13,400.00		4.80
DL	0.01	0.01	0.01		1.00

Enos - 10/03/2022: Oats, silage-soft dough

Enos Oat Silage

Sample and source description: Enos Oat Silage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	1,900.00	15,200.00		9.60
DL	0.01	0.01	0.01		1.00

Enos - 05/18/2023: Corn, silage

Enos Corn Silage

Sample and source description: Enos Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	1,500.00	22,000.00		6.30
DL	0.01	0.01	0.01		1.00

F/D1/D2 - 09/30/2022: Oats, silage-soft dough

Front/Dairy1/Dairy2 Oat Silage

Sample and source description: Front/Dairy1/Dairy2 Oat Silage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	3,800.00	23,800.00		10.10
DL	0.01	0.01	0.01		1.00

F/D1/D2 - 05/18/2023: Corn, silage

Front/Dairy1/Dairy2 Corn Silage

Sample and source description: Front/Dairy1/Dairy2 Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,000.00	1,800.00	14,000.00		6.50
DL	0.01	0.01	0.01		1.00

Gurr - 10/05/2022: Oats, silage-soft dough

Gurr Oat Silage

Sample and source description: Gurr Oat Silage

Sample date: 05/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	3,700.00	25,000.00		9.00
DL	0.01	0.01	0.01		1.00

Gurr - 05/25/2023: Corn, silage

Gurr Corn Silage

Sample and source description: Gurr Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,900.00	12,400.00		5.10
DL	0.01	0.01	0.01		1.00

Middle - 09/30/2022: Oats, silage-soft dough

Middle Field Oat Silage

Sample and source description: Middle Field Oat Silage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	3,000.00	25,100.00		9.80
DL	0.01	0.01	0.01		1.00

Middle - 05/18/2023: Corn, silage

Middle Field Corn Silage

Sample and source description: Middle Field Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	1,700.00	22,800.00		6.80
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

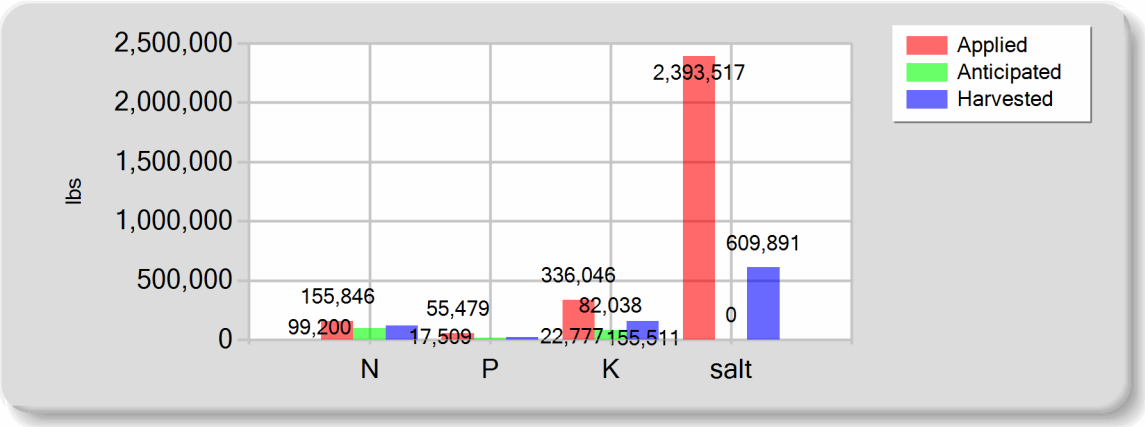
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

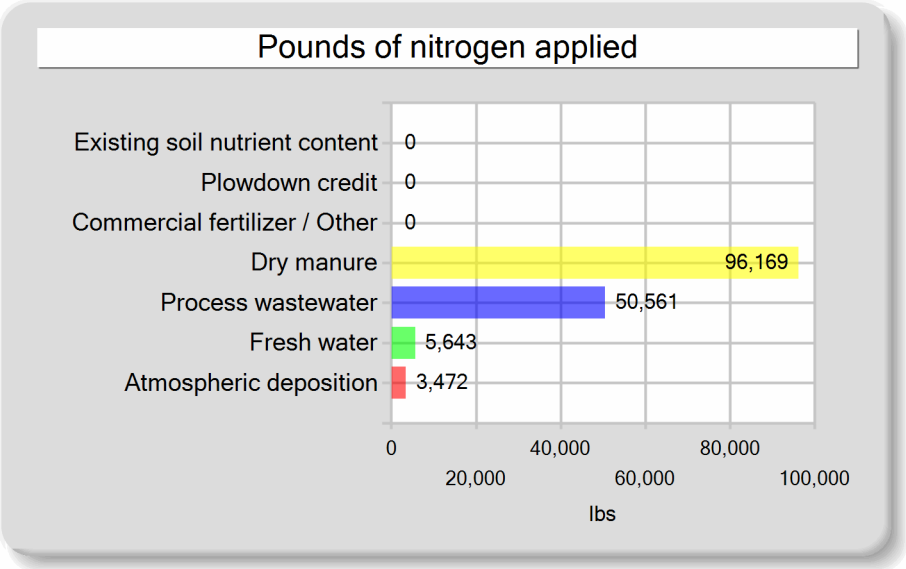
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

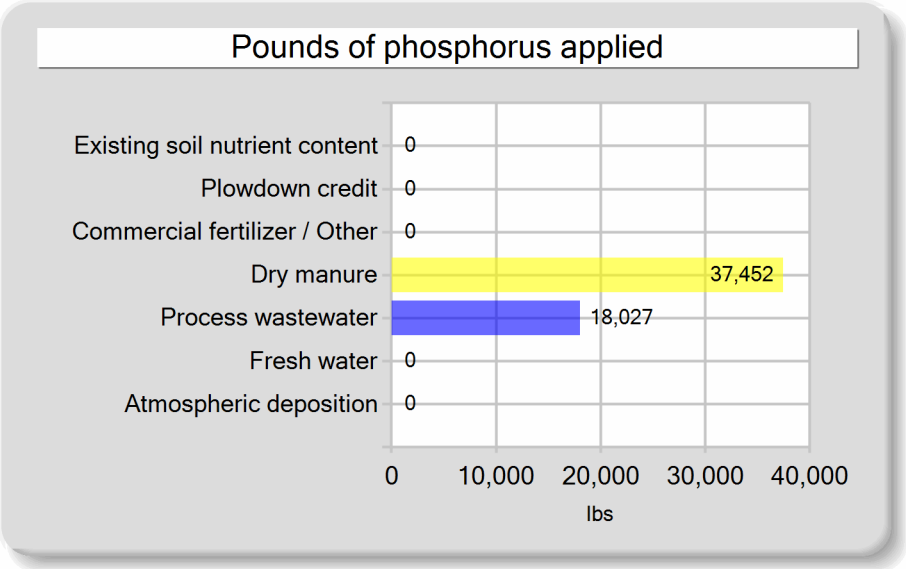
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	96,169.42	37,452.40	104,879.84	1,361,473.15
Process wastewater	50,561.05	18,026.90	231,165.97	990,440.77
Fresh water	5,643.06	0.00	0.00	41,603.55
Atmospheric deposition	3,472.00	0.00	0.00	0.00
Total nutrients applied	155,845.54	55,479.30	336,045.81	2,393,517.47
Anticipated crop nutrient removal	99,200.00	17,508.80	82,038.40	0.00
Actual crop nutrient removal	118,385.59	22,777.27	155,511.12	609,890.89
Nutrient balance	37,459.94	32,702.03	180,534.69	1,783,626.58
Applied to removed ratio	1.32	2.44	2.16	3.92

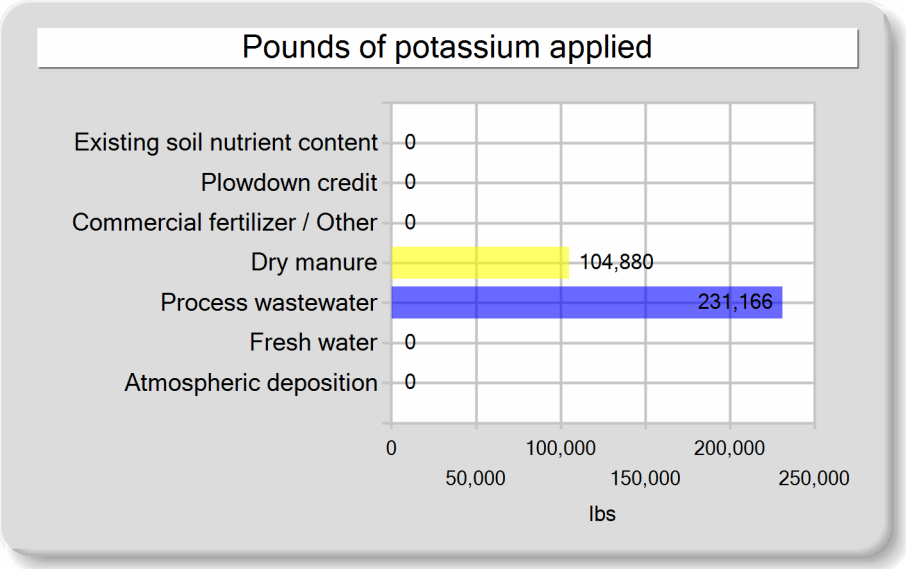
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

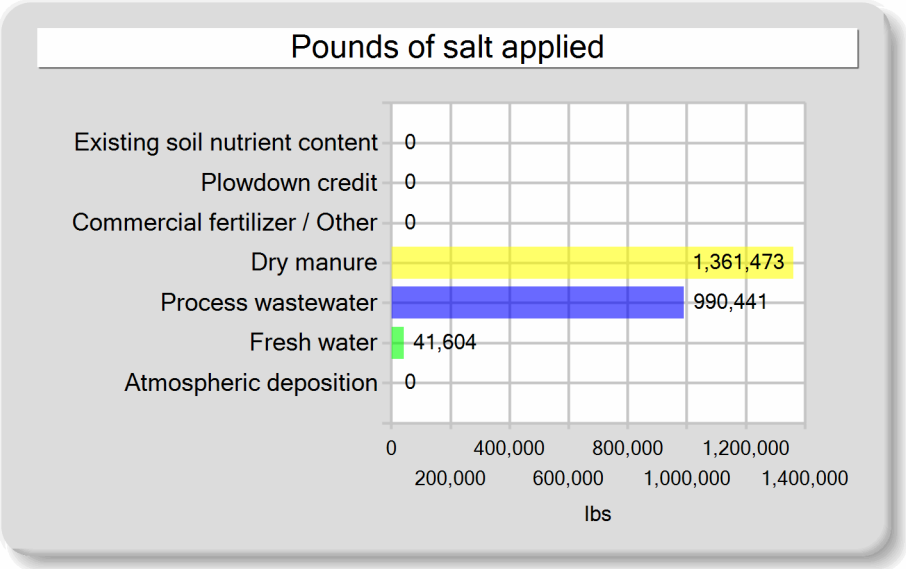


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Emanuel Z. Belo

SIGNATURE OF OWNER OF FACILITY

SIGNATURE OF OPERATOR OF FACILITY

Emanuel Belo

PRINT OR TYPE NAME

SAME AS OWNER

PRINT OR TYPE NAME

6/28/24

DATE

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

Belo Ranches
PO Box 635
Atwater, CA 95301

Account# 00-0021427
Account Manager: Ben Nydam
Submitted By: Emmanuel & Sonia Belo

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:40

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0876-01	DW - Barn	Well Water	M Dekeyrel		12/12/2023 11:08
23L0876-02	IW - Dairy	Well Water	M Dekeyrel		12/12/2023 11:12
23L0876-03	IW - Middle	Well Water	M Dekeyrel		12/12/2023 11:16
23L0876-04	IW - Enos	Well Water	M Dekeyrel		12/12/2023 11:20

Default Cooler Temperature on Receipt °C: 4.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Received: 12/13/2023 15:35
Reported: 12/21/2023 08:40

Sample Results

Sample: DW - Barn
23L0876-01 (Water)

Sampled: 12/12/2023 11:08
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.10**	mmhos/cm	0.01	1		12/18/23 11:52	SM 2510 B		BEL0676
Electrical Conductivity umhos	**1100**	umhos/cm	10.0	1		12/18/23 11:52	SM 2510 B		BEL0676
Ammonia (as N)	ND	mg/L	0.00	1		12/12/23 11:08	Field		BEL0559
Nitrate Nitrogen as NO3N	**10.2**	mg/L	0.1	1	10	12/17/23 01:53	EPA 300.0		BEL0664
Temperature	**25.0**	units	0.0	1		12/18/23 11:52	SM 4500-H+	H	BEL0676
pH	**7.3**	units	1.0	1		12/18/23 11:52	SM 4500-H+	H	BEL0676

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Received: 12/13/2023 15:35
Reported: 12/21/2023 08:40

Sample Results
(Continued)

Sample: IW - Dairy
23L0876-02 (Water)

Sampled: 12/12/2023 11:12
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.54**	mmhos/cm	0.01	1		12/18/23 11:54	SM 2510 B		BEL0676
Electrical Conductivity umhos	**542**	umhos/cm	10.0	1		12/18/23 11:54	SM 2510 B		BEL0676
Ammonia (as N)	ND	mg/L	0.00	1		12/12/23 11:12	Field		BEL0559
Nitrate Nitrogen as NO3N	**4.7**	mg/L	0.1	1	10	12/17/23 02:13	EPA 300.0		BEL0664
Temperature	**25.0**	units	0.0	1		12/18/23 11:54	SM 4500-H+	H	BEL0676
pH	**7.6**	units	1.0	1		12/18/23 11:54	SM 4500-H+	H	BEL0676

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Received: 12/13/2023 15:35
Reported: 12/21/2023 08:40

Sample Results
(Continued)

Sample: IW - Middle
23L0876-03 (Water)

Sampled: 12/12/2023 11:16
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.53**	mmhos/cm	0.01	1		12/18/23 11:55	SM 2510 B		BEL0676
Electrical Conductivity umhos	**534**	umhos/cm	10.0	1		12/18/23 11:55	SM 2510 B		BEL0676
Ammonia (as N)	ND	mg/L	0.00	1		12/12/23 11:16	Field		BEL0559
Nitrate Nitrogen as NO3N	**4.7**	mg/L	0.1	1	10	12/17/23 02:32	EPA 300.0		BEL0664
Temperature	**25.0**	units	0.0	1		12/18/23 11:55	SM 4500-H+	H	BEL0676
pH	**7.6**	units	1.0	1		12/18/23 11:55	SM 4500-H+	H	BEL0676

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Account# 00-0021427
Account Manager: Ben Nydam
Submitted By: Emmanuel & Sonia Belo

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:40

Sample Results
(Continued)

Sample: IW - Enos
23L0876-04 (Water)

Sampled: 12/12/2023 11:20
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.45**	mmhos/cm	0.01	1		12/18/23 11:57	SM 2510 B		BEL0676
Electrical Conductivity umhos	**448**	umhos/cm	10.0	1		12/18/23 11:57	SM 2510 B		BEL0676
Ammonia (as N)	ND	mg/L	0.00	1		12/12/23 11:20	Field		BEL0559
Nitrate Nitrogen as NO3N	**3.8**	mg/L	0.1	1	10	12/17/23 02:52	EPA 300.0		BEL0664
Temperature	**25.0**	units	0.0	1		12/18/23 11:57	SM 4500-H+	H	BEL0676
pH	**7.6**	units	1.0	1		12/18/23 11:57	SM 4500-H+	H	BEL0676

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Account# 00-0021427
Account Manager: Ben Nydam
Submitted By: Emmanuel & Sonia Belo

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:40

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0664										
Blank (BEL0664-BLK1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0664-BLK2)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0664-BLK3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0664-BLK4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0664-BLK5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0664-BS1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
LCS (BEL0664-BS2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
LCS (BEL0664-BS3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEL0664-BS4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEL0664-DUP1)		**Source: 23L0865-01**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	3.8		0.1	mg/L		3.7			0.802	10
Duplicate (BEL0664-DUP2)		**Source: 23L0871-04**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L		5.2			0.155	10
Duplicate (BEL0664-DUP3)		**Source: 23L0984-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.0		0.1	mg/L		1.0			0.305	10
Duplicate (BEL0664-DUP4)		**Source: 23L0984-03**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			1.68	10
Matrix Spike (BEL0664-MS1)		**Source: 23L0865-01**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	8.8		0.1	mg/L	5.000	3.7	101	90-110		
Matrix Spike (BEL0664-MS2)		**Source: 23L0871-04**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2		0.1	mg/L	5.000	5.2	102	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0664 (Continued)										
Matrix Spike (BEL0664-MS3)			**Source: 23L0984-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.2		0.1	mg/L	5.000	1.0	104	90-110		
Matrix Spike (BEL0664-MS4)			**Source: 23L0984-03**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.5		0.1	mg/L	5.000	1.3	103	90-110		
Reference (BEL0664-SRM1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0664-SRM2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0664-SRM3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0664-SRM4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0664-SRM5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0676										
Blank (BEL0676-BLK1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.2		1.0	units						
Blank (BEL0676-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0676-BLK3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0676-DUP1)		**Source: 23L0871-05**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	0.60		0.01	mmhos/cm		0.61			0.979	10
pH	7.6		1.0	units		7.5			1.85	10
Electrical Conductivity umhos	599		10.0	umhos/cm		605			0.979	10
Duplicate (BEL0676-DUP2)		**Source: 23L0882-03**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	0.90		0.01	mmhos/cm		0.90			0.0111	10
Electrical Conductivity umhos	900		10.0	umhos/cm		900			0.0111	10
pH	7.9		1.0	units		7.9			0.127	10
Reference (BEL0676-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	431			umhos/cm	426.0		101	90-110		
Reference (BEL0676-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.6			units	7.520		100	67021-101.32		
Reference (BEL0676-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0676-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEL0676-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0676 (Continued)										
Reference (BEL0676-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0676-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0676-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0676-SRM8)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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12/13/23 15:35 23L0876

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21427 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Belo Ranches
Address: P.O. Box 635
City, State, Zip: Atwater, CA 95301
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Emanuel & Sonia Belo - (209) 358-3144

Ranch _____

Date sampled: 12/12/23

Sampled by: mDekuyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[A] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW - Barn	Sampled From:	12/12/23	1108A	Ø	4.4
2. IW - Dairy	Sampled From:	12/12/23	1112A	Ø	4.4
3. IW - Middle	Sampled From:	12/12/23	1116A	Ø	5.1
4. IW - Enos	Sampled From:	12/12/23	1120A	Ø	5.5
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)		Relinquished (Date/Time)	
First	mDekuyrel	OASInc	12/12/23	1120A	12/13/23	11:20A
Second	JLF	DLV	12-13-23	11:00	12-13-23	3:01
Third						
Fourth	Jg	DL2	12/13/23	15:35		

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1	1						
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/13/23 15:35 23L0876

SEND OUT

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Belo Ranches
PO Box 635
Atwater CA 95301
00-0021427 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0876
Sampled By: M Dekeyrel
Requested By: Emmanuel & Sonia Belo
Submitted Date: 12/13/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - Barn	12/12/23	11:08	1100	1.10	10.2	ND		7.3
2	IW - Dairy	12/12/23	11:12	542	0.54	4.7	ND		7.6
3	IW - Middle	12/12/23	11:16	534	0.53	4.7	ND		7.6
4	IW - Enos	12/12/23	11:20	448	0.45	3.8	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Belo Ranches
PO Box 635
Atwater CA 95301
00-0021427 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0876
Sampled By: M Dekeyrel
Requested By: Emmanuel & Sonia Belo
Submitted Date: 12/13/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	**EC** µmhos/cm	**EC** mmhos/cm	**NO_3-N** mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - Barn	12/12/23	11:08	**1100**	**1.10**	**10.2**	27.85	2.32
2	IW - Dairy	12/12/23	11:12	542	0.54	**4.7**	12.83	1.07
3	IW - Middle	12/12/23	11:16	534	0.53	**4.7**	12.83	1.07
4	IW - Enos	12/12/23	11:20	448	0.45	**3.8**	10.37	0.86

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.